RECOMMENDAI

An AI-Based Hybrid E-Commerce Recommendation System

PROJECT REPORT

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Academic Project Report

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Abstract

RecommendAI is an AI-based e-commerce recommendation system designed to provide personalized product suggestions using a combination of user behaviour, product information, popularity signals, and shopping patterns. The project description identifies a hybrid recommendation pipeline combining content-based filtering, collaborative filtering, time-decayed trending signals, and market-basket association rules. A ranking stage combines recommendation scores and applies constraints such as stock availability and exclusion of previously purchased products. Maximal Marginal Relevance (MMR) is used to improve diversity in the final list.

The system also describes explainable recommendation badges, a cold-start onboarding flow, product bundles, a shopping cart interface, demo personas, and an administrator console for monitoring interactions and reviewing information-retrieval metrics. The intended contribution is to demonstrate how several recommendation approaches can be combined into one interactive application. The reported metrics in the project description should be treated as illustrative until reproduced from the actual dataset and evaluation code.

1. Introduction

E-commerce platforms offer large product catalogues, making it difficult for shoppers to find relevant products efficiently. Recommendation systems address this problem by ranking products that may match a user's interests, current activity, or shopping context.

RecommendAI is presented as a hybrid recommender rather than a single-algorithm system. It combines product similarity, user–item interaction patterns, current popularity, and co-purchase relationships. Its interface is designed to demonstrate personalized feeds, explanations, bundles, cold-start onboarding, and administrative evaluation.

2. Problem Statement

A basic online store may display products through category filters, fixed sorting, or general popularity. These approaches may not adequately reflect individual preferences, may repeat similar items, and may struggle when a user has little or no interaction history.

The project aims to create a recommendation pipeline that combines multiple signals, handles new users through stated interests, filters unavailable products, and introduces diversity into the ranked results. It also aims to make recommendation decisions easier to inspect through explanation labels and evaluation summaries.

3. Objectives

• Build a hybrid recommendation pipeline that combines content-based and collaborative signals.

• Use product popularity with time decay to give more weight to recent activity.

• Identify related products through market-basket association rules.

• Combine model signals into a final ranking and improve list diversity using MMR.

• Provide a cold-start flow based on user-selected interests.

• Explain recommendation reasons using model-derived attribution labels.

• Demonstrate bundles, product details, search, filters, cart operations, and a simulated checkout.

• Present interaction telemetry and information-retrieval metrics in an administrative console.

4. Scope and System Overview

The described application contains a customer-facing shopping interface and an administrative/demo interface. Demo personas represent different shopping interests, while a cold-start guest illustrates the experience of a user

with no prior history. The admin persona is described as a route to telemetry, evaluation tables, and model-operations controls.

The report is based on the supplied project description. The uploaded ZIP was inspected but contained only a Windows shortcut file (recommendai.lnk), not the source-code files. Therefore, implementation details below document the stated design and are not an independent line-by-line code audit.

5. Recommendation Methodology

5.1 Content-Based Filtering: Product titles, brands, categories, tags, and specifications are described as text inputs for TF-IDF vectorization. Cosine similarity is then used to compare product representations with a profile derived from the user's browsing or interaction history. This can recommend items with similar attributes even when interaction data is limited.

5.2 Collaborative Filtering: The design describes implicit-feedback Alternating Least Squares (ALS) matrix factorization. Views, clicks, add-to-cart events, and purchases are assigned different strengths to represent varying levels of interest. The model learns latent user and item factors from these interactions. Actual performance depends on the dataset, preprocessing, training configuration, and evaluation protocol.

5.3 Time-Decayed Trending: Recent product activity is given more influence than older activity through an exponential decay function. The stated design uses a seven-day half-life. This can help the system react to changing trends instead of relying only on lifetime popularity.

5.4 Market-Basket Association Rules: FP-Growth is described as the method for discovering products that frequently occur together in transactions. Support measures how often an itemset occurs; confidence estimates the conditional frequency of a consequent given an antecedent; lift compares the observed co-occurrence with the co-occurrence expected under independence. The description proposes using high-lift rules to support related-product bundles.

5.5 Score Fusion: The stated baseline score combines collaborative, content, and popularity signals using weights of 0.45, 0.35, and 0.20 respectively. For these weights to behave as intended, the component scores should be normalized or calibrated to comparable ranges. The described fusion does not explicitly include the FP-Growth score in the displayed formula; association rules are presented primarily for bundles.

5.6 Re-ranking with MMR: Maximal Marginal Relevance balances relevance with dissimilarity to items already selected. It can reduce repetitive lists, such as recommendations dominated by a single product type. The stated design uses a lambda value of 0.75; the precise effect depends on the MMR formulation and score normalization.

5.7 Business Constraints: The recommendation feed is described as excluding products with no stock and products already purchased by the user. These are useful eligibility rules, although the system should distinguish purchased-item exclusion from legitimate repeat-purchase scenarios where replenishment is appropriate.

6. Cold-Start Handling and Explainability

Cold-start onboarding asks a new user to select at least two areas of interest, such as gaming, audio, or computing. The selected interests are used to construct an initial preference profile so the system can produce a personalized first-session feed without historical clicks or purchases.

Recommendation cards display reason labels such as collaborative filtering, content similarity, trending popularity, or selected category interest. To keep explanations trustworthy, the displayed reason should be generated from the actual score components and decision path for that item. If several signals contribute, the interface should avoid presenting a single reason as mathematically dominant unless this is verified.

7. Product Bundles and Shopping Experience

The described bundle feature uses association patterns to suggest complementary products, for example a laptop and dock or a keyboard and mouse. A bundle action can add both items to the cart and apply a proposed discount.

Discounts, tax, totals, stock checks, and checkout behaviour should be validated against the application's actual business logic before being represented as production-ready.

The shopping interface is described as including product detail views, price and discount information, specifications, stock status, search, category filters, quantity adjustments, a cart drawer, tax calculation, and simulated checkout. The visual design uses a dark glassmorphism style, neon accents, and an interactive particle background.

8. System Architecture

At a conceptual level, the pipeline is: user interactions and product catalogue → feature/profile construction → candidate generation from content, collaborative, and trending sources → score normalization and fusion → eligibility constraints → diversity re-ranking → explanation labels → recommendation feed.

Association-rule mining supports a parallel bundle path: transaction history → frequent itemsets and association rules → related-product/bundle suggestions. An administrative path collects interaction events and presents monitoring and offline-evaluation summaries.

Figure description for implementation: a system architecture diagram should show these two paths and the shared product catalogue. The precise deployment architecture, framework versions, API boundaries, persistence layer, and model-training schedule cannot be confirmed from the shortcut-only archive.

9. Evaluation Approach

The project description lists Precision@5, Recall@5, NDCG@10, MAP@10, HitRate@10, catalogue coverage, and diversity as evaluation measures. These metrics assess different properties: precision measures the proportion of recommended items that are relevant; recall measures how many relevant items are retrieved; NDCG rewards relevant items near the top; MAP averages precision at relevant ranks; hit rate checks whether at least one relevant item appears; coverage measures catalogue exposure; and diversity measures dissimilarity among recommendations.

The description includes a comparison table with a reported hybrid-system Precision@5 of 25.4%, Recall@5 of 41.2%, NDCG@10 of 0.468, MAP@10 of 0.364, HitRate@10 of 59.5%, Coverage of 84.6%, and Diversity of 0.672. These values are reproduced here as project-provided figures, not independently verified results. They should only be presented as experimental findings after confirming the dataset, train/test split, negative sampling, metric implementation, and reproducibility of the evaluation code.

Recommended evaluation procedure: use a temporal holdout split; train models only on interactions before the cutoff; evaluate on later relevant interactions; compare against random, popularity, content-based, and collaborative baselines; report the number of users and items, interaction counts, candidate-generation rules, and metric definitions. Include ablation tests to measure the effect of each component, especially hybrid fusion and MMR.

Metric	Reported value	Status
Precision@5	25.4%	Project-provided; verify
Recall@5	41.2%	Project-provided; verify
NDCG@10	0.468	Project-provided; verify
MAP@10	0.364	Project-provided; verify
HitRate@10	59.5%	Project-provided; verify
Coverage	84.6%	Project-provided; verify
Diversity	0.672	Project-provided; verify

Note: figures are included from the project description and have not been independently reproduced.

10. Testing Plan

Functional tests should cover onboarding for a new user, recommendation generation for each demo persona, out-of-stock filtering, purchased-item exclusion, search and category filtering, bundle generation, cart quantity changes, totals and discount calculations, and explanation consistency.

Model tests should verify that score components are finite and normalized, empty-history users receive sensible fallback recommendations, repeated requests are deterministic where expected, MMR reduces redundant items under controlled examples, and association rules meet minimum support/confidence thresholds.

Evaluation tests should check metric implementations using small hand-calculated examples and ensure that no test interactions leak into training. Administrative metrics should be clearly labelled as live, offline, simulated, or sample data depending on their actual source.

11. Limitations and Future Enhancements

Hybrid recommendation quality depends on data quality, sufficient interaction history, and careful score calibration. Synthetic demo personas can demonstrate user journeys but cannot establish real-world effectiveness. Offline ranking metrics do not prove higher conversion, revenue, or user satisfaction. A visually live telemetry feed should not be described as production telemetry unless it is connected to a genuine event stream.

Potential future work includes user-controlled recommendation objectives, session-intent detection, feedback-driven updates, counterfactual explanations, filter-bubble monitoring, budget and other hard constraints, seller/brand exposure analysis, and online experiments. Any feature not currently implemented should be labelled as proposed future work rather than a completed capability.

12. Conclusion

RecommendAI brings together several established recommendation techniques in a single e-commerce demonstration. Its stated design combines TF-IDF content similarity, implicit-feedback ALS, time-decayed popularity, association-rule bundles, weighted ranking, business constraints, MMR diversity, cold-start onboarding, and explanation labels. The admin console is intended to make ranking quality and catalogue-level behaviour easier to inspect.

The main contribution is the integration of multiple recommendation signals with diversity, cold-start support, and user-facing explanations. The next important step is to verify each stated component against the source code and reproduce the evaluation results from a documented dataset and protocol.

References

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